

Unit 5: Earth–Sun Relationships — Reasons for the Seasons

Reading: pages 66–75 · 20 points · 15 minutes · Closed book

Name _____

Date _____

Score / 20

Answer every question. For multiple choice, circle the single best answer. Write short answers in complete sentences.

Part A · Multiple Choice

6 questions × 2 points = 12 points

1. In 1543, who proposed that Earth and the other planets make regular revolutions around the Sun, and that Earth rotates once a day on its axis?
 - a) Edwin Hubble
 - b) James Hutton
 - c) Galileo Galilei
 - d) Nicolaus Copernicus
2. Earth's seasons are caused by
 - a) Earth being physically closer to the Sun in summer
 - b) the 23.5° tilt of Earth's axis combined with Earth's revolution around the Sun
 - c) regular changes in the Sun's energy output
 - d) the greenhouse effect
3. At the Northern Hemisphere's summer solstice, the Sun's rays strike Earth most directly along
 - a) the Tropic of Cancer, 23.5° N
 - b) the Tropic of Capricorn, 23.5° S
 - c) the Arctic Circle, 66.5° N
 - d) the equator
4. Which type of ultraviolet radiation does not reach the planet's surface at all?
 - a) UVA
 - b) UVB
 - c) UVC
 - d) Infrared
5. Compared with one molecule of carbon dioxide, one molecule of methane traps about
 - a) the same amount of heat
 - b) 23 times as much heat
 - c) 250 times as much heat
 - d) 10,600 times as much heat

6. On an equinox,
- a) the Sun's rays shine most directly at the equator and daylight and nighttime hours are equal
 - b) the North Pole points directly toward the Sun
 - c) Earth reaches its closest point to the Sun
 - d) the Sun never rises above the horizon at either pole

Part B · True or False

2 questions × 1 point = 2 points

7. About 44% of the solar radiation reaching Earth is in visible light wavelengths.

TRUE / FALSE

8. Greenhouse gases are entirely a product of human activity.

TRUE / FALSE

Part C · Short Answer

2 questions × 3 points = 6 points

9. Explain why the Northern Hemisphere is warmer in June than in December. Your answer should refer both to the *directness* of the Sun's rays and to the *length* of daylight.

10. What is an *analemma*, and which two features of Earth's motion give it its figure-eight shape?

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